

A berry thought-provoking idea: the potential role of plant polyphenols in the treatment of age-related cognitive disorders.

Today, tens of millions of elderly individuals worldwide suffer from dementia. While the pathogenesis of dementia is complex and incompletely understood, it may be, at least to a certain extent, the consequence of systemic vascular pathology. The metabolic syndrome and its individual components induce a proinflammatory state that damages blood vessels. This condition of chronic inflammation may damage the vasculature of the brain or be directly neurotoxic. Associations have been established between the metabolic syndrome, its constituents and dementia. A relationship has also been observed between certain dietary factors, such as constituents of the 'Mediterranean diet', and the metabolic syndrome; similar associations have been noted between these dietary factors and dementia. Fruit juices and extracts are under investigation as treatments for cognitive impairment. Blueberry, strawberry, blackberry, grape and plum juices or extracts have been successfully tested in cognitively impaired rodents. Published trials of the benefits of grape and blueberry juice in the treatment of small numbers of cognitively impaired persons have recently appeared. The benefits of fruit products are thought to be a result of its polyphenol content. A grape polyphenol found in grapes, resveratrol, now being studied in humans, and one in grapes and blueberries, pterostilbene, have been found to improve cognition in rodents. In the design of future human trials, one ought to consider the poor bioavailability of these products, the possible need to initiate the experimental therapy long before the onset of symptoms, and currently limited knowledge about the appropriate form (e.g. juice, powder or individual polyphenol) of treatment.